Yufeng (Alex) Xia

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Education

- **University of Edinburgh** — Edinburgh, UK
  MSc by Research in Computer Science, supervised by Prof. ***Mahesh K. Marina*** — *Sep 2025 – Present*
- **University of Edinburgh** — Edinburgh, UK
  Bachelor of Science in Computer Science; Grade: ***First Class (Hons)*** — *Sep 2021 – May 2025*

Research Experience

- **University of Edinburgh – Research Assistant** — ICSA / NetSys Group
  2024 – Present
  - **LLVM Compiler Framework**: Developed CoSenseAQ, an LLVM-based auto-quantization framework leveraging sensor specifications for code size and performance optimization on embedded ARM systems.
  - **Scalable Edge Emulator**: Participated in developing Morphling, a measurement-driven emulation platform supporting 1800+ heterogeneous edge devices for distributed ML training evaluation.
  - **AI-RAN Infrastructure**: Contributed PHY experiments and GPU profiling for Weaver, a system exploring opportunistic foundation model training on GPU-accelerated RAN infrastructure.
  - **RAN Emulation at Scale**: Contributed to system deployment, evaluation, and paper writing for EmuRAN, a cloud-based RAN emulation system supporting 10,000+ devices and base stations.

Technical Skills

- C/C++, Python, LLVM IR, Kubernetes, Open5GS, OAI, Linux, Docker

Research Interests

- Networking and ML Systems, AI-RAN, Digital Twins, Distributed Edge Intelligence

Selected Publications

- **EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** — MobiCom 2026, Conditionally Accepted
  - Built a cloud-based RAN emulation system capable of emulating 10,000+ mobile devices and base stations.
- **Morphling: Emulator for Distributed Machine Learning at the Edge** — EdgeSys 2026
  - Developed a measurement-driven emulator for evaluating distributed ML training on heterogeneous edge devices at scale.
- **WASP: Foundation Model Training System for the Edge** — Under Submission
  - Designed a foundation model training system for resource-constrained edge environments.
- **Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** — Under Submission
  - Enabled opportunistic foundation model training on GPU-accelerated AI-RAN infrastructure.
- **CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** — Under Submission
  - Developed an LLVM-based auto-quantization framework leveraging sensor specifications for embedded systems.

Awards

- **Edinburgh Award**: Recognition of professional growth and invaluable experiences gained as a Research Assistant.